

Carbs are king when it comes to reaching your potential as a swim athlete. Carbs are energy (aka fast swimming!); the breakdown of carbs generates a lot of ATP, the body's energy currency. Carbs are brain fuel; they help you focus in school and react off the blocks. Carbs are muscle fuel; they help you build/maintain lean muscle. Many foods provide carbohydrates for the swim athlete to perform at their highest level. Both unprocessed, whole food sources and processed sources can be used as part of a healthy, high performing fueling plan.

A carbohydrate can be simple (short-chain) or complex (long-chain). Simple carbs are also known as “fast” carbs that digest rapidly and easily, while complex carbs are also known as “slow” carbs that provide sustained energy. Check out the *ABC's of Nutrition* for some food examples of these.

Simple carbohydrates, as mentioned, are very easy for the body to digest and contain monosaccharides and/or disaccharides. Glucose, for example, is a monosaccharide which is stored as glycogen (a long chain of glucose molecules) in the muscles and liver.

Complex carbs can be maltodextrins (a combo of 6-20 glucose molecules), starches, fibers, or glycogen.

Whether simple or complex, the end result of carbohydrate breakdown is mostly glucose (sugar). This glucose is used by our cells and starts the process of creating ATP for energy.

Sports drinks, gummies/chews, rice krispies, honey, maple syrup, crackers, pretzels, fruit bars, dried mango, bananas and dates are all comprised of short-chain carbs. These options are easy to digest and quick for the body to use as fuel. While a piece of fruit (like a banana, date) has naturally occurring vitamins/minerals, these options also contain fiber which may not be well-tolerated by many athletes in the ~30-60 minutes prior to training/racing. Every athlete has an individual response and should be practicing their fueling strategy just like they practice technique in the pool.

On the other hand, whole grains like quinoa, oats, and brown rice, legumes, and high-fiber fruits/veggies contain long-chain carbohydrates that take longer to digest and are not a good source of quick fuel in the ~30-60 minutes before training/racing. Rather, these options are great as part of larger meals that are spaced further away from training/racing in order to give the body ample time to digest.

SWIMMER'S GUIDE TO CARBS



Quick Carbs
(30-60 minutes before
training)



Slower Carbs (2+ hours
before training/racing)



Eliminating processed foods altogether may put an athlete at higher risk for entering a state of relative energy deficiency. RED-S (relative energy deficiency in sport) has many serious complications, both in the short and long term. RED-S can cause symptoms like impaired growth and development, loss of menstrual cycle in females, poor recovery, low energy, decreased performance, impaired GI health, and poor mental health, to name a few.

Rather than eliminating processed foods altogether, we recommend strategically utilizing them as part of a nutrient dense fueling plan. Processed foods fall on a spectrum (milk, for example, is processed) and a small amount utilized intentionally has been shown to benefit performance and health of the growing athlete. Both whole, unprocessed foods as well as processed foods can be used to the swimmer's advantage and put them ahead of the competition.